

COURSE TITLE : **BIOMATERIALS AND PROSTHETIC DEVICES**
COURSE CODE : **5243**
COURSE CATEGORY : **E**
PERIODS/WEEK : **4**
PERIODS/SEMESTER : **52**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Bio material and its applications	13
2	Polymers and Composites	13
3	Prosthetic and orthotic devices	13
4	Measurement of biomechanics	13
	TOTAL	52

Course General Outcome:

MODULE	G.O.	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE
1	1	To understand the biomaterial and its characteristics
	2	To distinguish metals and ceramics
2	3	To understand polymers, composites and their applications
3	4	To understand prosthetic devices
	5	To understand orthotic devices
4	6	To understand Tonometers
	7	To understand locomotion monitoring devices

On completion of the study the student will be able

MODULE I: BIO MATERIAL AND ITS APPLICATIONS

1.1.0 To understand the biomaterial and its characteristics

- 1.1.1 To define the bio materials
- 1.1.2 To list the types of Bio materials
- 1.1.3 To state the requirements of bio materials
- 1.1.4 To define bio compatibility
- 1.1.5 To explain the material response to bio materials
- 1.1.6 To explain host response to bio materials
- 1.1.7 To explain tissue response to bio materials
- 1.1.8 To classify biomaterials, metals, ceramics, polymers and composites.

1.2.0 To distinguish metals and ceramics

- 1.2.1 To explain the composition and properties of stainless steel alloys
- 1.2.2 To explain the composition and properties of co based alloys
- 1.2.3 To describe the fracture fixation devices
- 1.2.4 To describe the structure and properties of different types of ceramic implants
- 1.2.5 To explain the dental implants

MODULE II: POLYMERS AND COMPOSITES

2.1.0 To understand polymers, composites and their applications

- 2.1.1 To describe the polymeric implant materials
- 2.1.2 To explain the structure and properties of polyolefins
- 2.1.3 To explain the structure and properties of polyamides
- 2.1.4 To explain the structure and properties of acrylic polymers
- 2.1.5 To describe natural rubber and synthetic rubber
- 2.1.6 To describe thermoplastics
- 2.1.7 To explain the Bio degradable polymer and its application
- 2.1.8 To describe percutaneous and soft tissue replacements and vascular implants
- 2.1.9 To explain properties of composites.
- 2.1.10 To list applications of composites.

MODULE III: PROSTHETIC AND ORTHOTIC DEVICES

3.1.0 To understand prosthetic devices

- 3.1.1 To define prosthetic devices
- 3.1.2 To differentiate homograft and heterograft
- 3.1.3 To explain artificial heart valves and its requirements
- 3.1.4 To explain types of artificial heart valves
- 3.1.5 To explain hip prosthesis
- 3.1.6 To explain the biological requirements of hip prosthesis
- 3.1.7 To explain knee joint prosthesis

3.2.0 To understand orthotic devices

- 3.2.1 To define orthotic devices
- 3.2.2 To differentiate between crutches and canes.
- 3.2.3 To list different types of crutches and canes
- 3.2.4 To explain wheel chairs
- 3.2.5 To define Scoliosis
- 3.2.6 To list types of correction procedures for scoliosis
- 3.2.7 To explain exercise procedures for the correction of scoliosis
- 3.2.8 To explain Milwaukee brace for the correction of scoliosis
- 3.2.9 To explain Dwyer's technique for the correction of scoliosis

MODULE IV: MEASUREMENT OF BIOMECHANICS

4.1.0 To understand Tonometers

- 4.1.1 To define glaucoma.
- 4.1.2 To list the different types of tonometers
- 4.1.3 To explain Goldmanns Applanation tonometer.
- 4.1.4 To explain Schiotz impression tonometer
- 4.1.5 To explain MacKayMarg indentation tonometer

4.2.0 To understand locomotion monitoring devices

4.2.1 To explain the human locomotion monitoring system

4.2.2 To define gait, swing phase and stance phase

4.2.3 To explain the foot switches for gait analysis

4.2.4 To describe foot pressure measurement.

4.2.5 To explain pedobarograph

CONTENTS

MODULE I BIOMATERIAL AND ITS CHARACTERISTICS

Definition of bio materials- types and requirements- Definition of bio compatibility- Material response , host response and tissue response to bio materials- Classification of biomaterials, metals, ceramics, polymers and composites.

Metals and ceramics

Composition and properties of stainless steel alloys and Co based alloys - Fracture fixation devices - Structure and properties of different types of ceramic implants - Dental implants

MODULE II POLYMERS, COMPOSITES AND THEIR APPLICATIONS

Polymeric implant materials-- structure and properties of polyolefins, polyamides, acrylic polymers - Natural rubber and synthetic rubber – Thermoplastics - Bio degradable polymer and its application- Percutaneous and soft tissue replacements and vascular implants-Properties and applications of composites.

MODULE III PROSTHETIC DEVICES

Definition of prosthetic devices - Homograft and heterograft- Artificial heart valves, types and its requirements - Hip prosthesis and its biological requirements - Knee joint prosthesis.

Orthotic devices

Definition of orthotic devices - Crutches and canes and their types- wheel chairs- Definition of Scoliosis- Correction procedures for scoliosis -Exercise procedures for the correction of scoliosis

-Milwaukee brace- Dwyer's technique.

MODULE IV TONOMETERS

Definition of glaucoma – Goldmanns Applanation tonometer - Schiotz impression tonometer- MacKay-Marg indentation tonometer.

Locomotion monitoring devices

Human locomotion monitoring system- Definition of gait, swing and stance phase-Foot switches for gait analysis -Foot pressure measurement - Pedobarograph

TEXT BOOK

1. Biomaterials – Sujatha v Butt- Alpha Science International, 2005
2. Biomechanics of medical devices-DN Ghista- Taylor & Francis, 1981

REFERENCES

1. Biomedical science and energy- J B Park, Plenum press 1984
2. Biomedical polymers- A Rembaum and M Shen- M. Dekker, 1971
3. Encyclopaedia of medical devices and instruments-John G Webster- Wiley, 1988